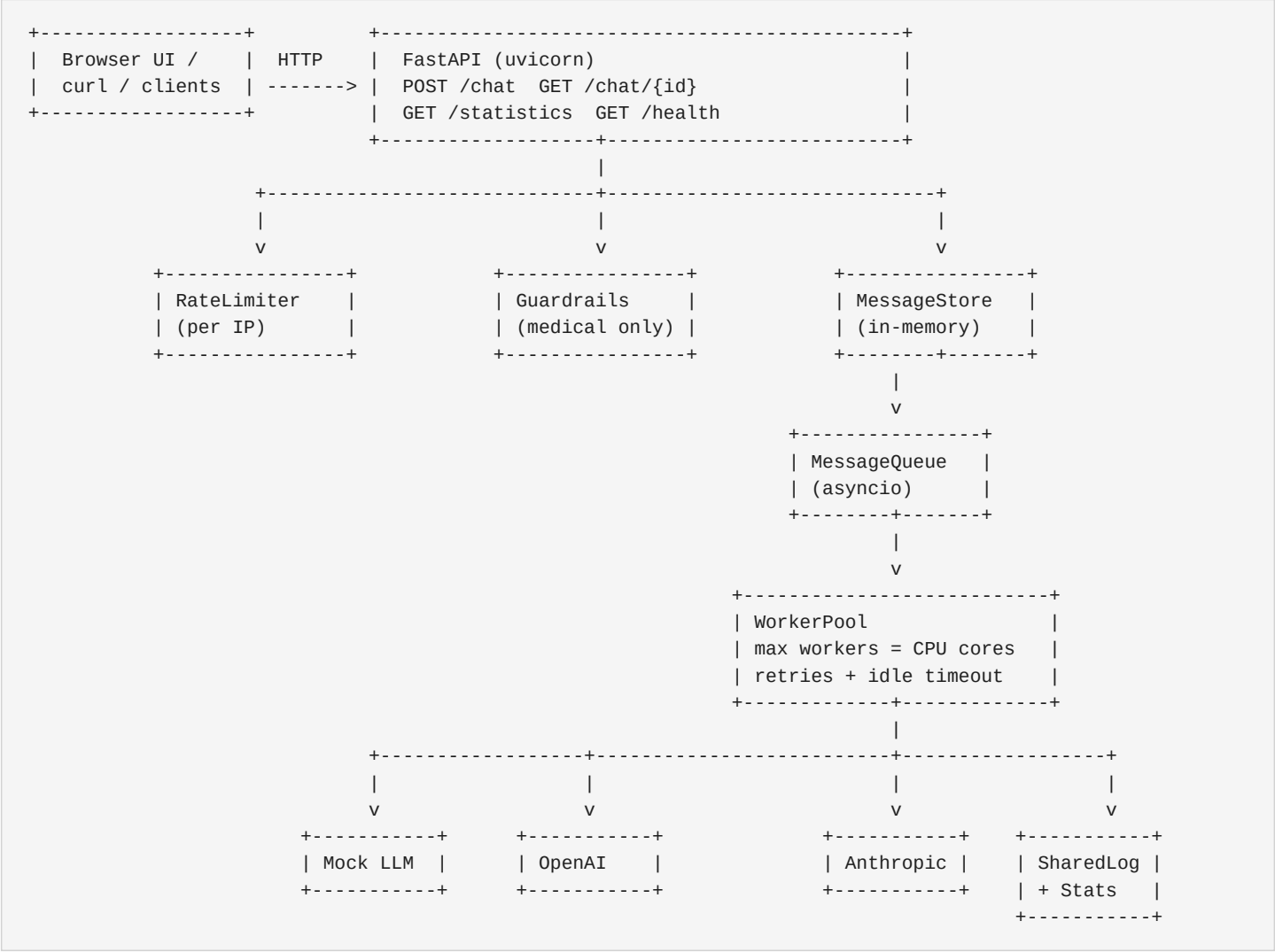


Medical Expert AI Chat

System Architecture Overview

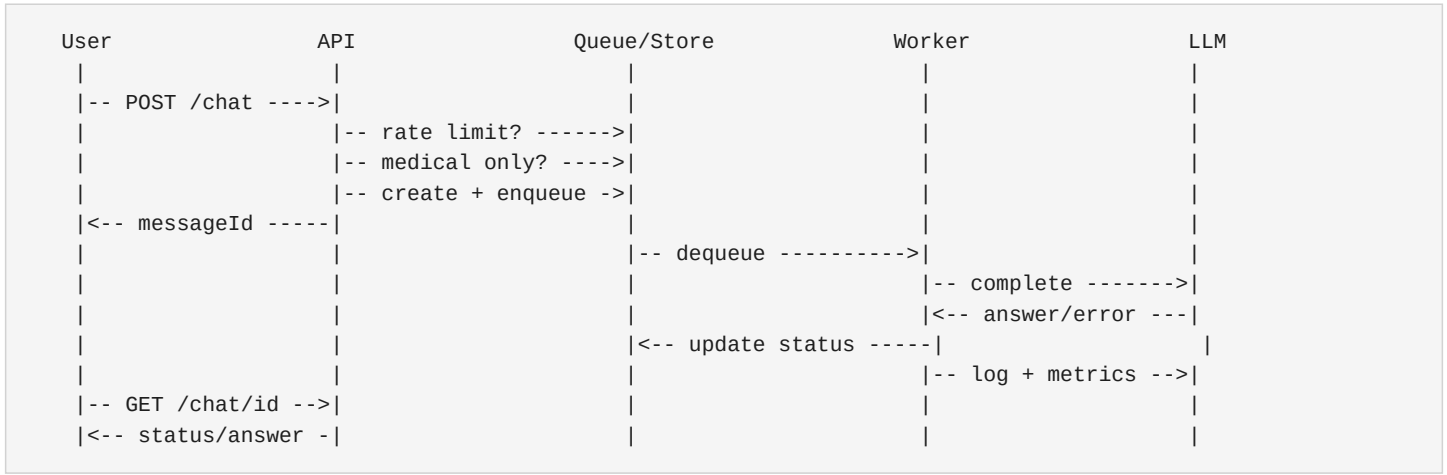
1. High-level architecture

Self-contained Python 3.13+ service: FastAPI HTTP API, async worker pool, LLM providers (mock / OpenAI / Anthropic), shared file logging, statistics, and a simple polling UI. Abuse controls: per-IP rate limiting and medical-only guardrails.



2. Request lifecycle

Submit returns immediately with messageId. Workers process asynchronously. Clients poll GET /chat/{messageId} until completed or failed.



3. Module map

Item	Description
main.py	App factory, lifespan, static UI
api.py	HTTP routes
config.py	Env-based Settings (pydantic-settings)
guardrails.py	Reject non-medical / abusive input
rate_limit.py	Per-IP sliding-window limiter
storage.py	Message store, queue, shared log
worker_pool.py	Dispatcher, workers, retries, idle exit
statistics.py	Metrics aggregation
llm/	Provider clients + factory
domain.py	ChatMessage and log entry models
static/	Simple frontend (poll + stats)

4. Design choices

Item	Description
Async processing	API returns immediately; workers run in background
Concurrency	asyncio queue + tasks; max = CPU cores
State	In-memory MessageStore (not durable across restart)
Logs	Append-only JSON lines file (thread-safe lock)
Config	Environment variables via Settings
Abuse control	Rate limit + medical-only filter + LLM prompt
Deployment	Docker / uvicorn on port 8000; GET /health
Python	Requires Python 3.13+

5. Data status lifecycle

```
pending --> processing --> completed
                \--> failed
```

Each transition updates MessageStore.
Completed/failed also append SharedLog and update StatisticsCollector.

6. HTTP API

Item	Description
POST /chat	Submit question -> { messageId }
GET /chat/{id}	processing completed+answer failed+error
GET /statistics	Queue, workers, retries, tokens, timings

GET /health	Deployment health check
GET /	Simple chat UI

7. Out of scope / optional

Auth, real medical validation, and polished UI are out of scope. Optional bonuses not implemented: continuous chat history, persistence across restarts, and streaming responses.